Spectrally constrained stochastic factorization for low-dose CT

1 Low-dose CT reconstruction model

We reconstruct the X-ray attenuation of a two-dimensional slice from low-dose measurements taken at multiple angles. The unknown image X represents attenuation on a 512×512 grid.

What the scanner measures. An attenuation image describes how strongly the material at each location reduces the transmitted X-ray beam per unit distance. At one angular position, called a *view*, an X-ray source illuminates a line of detector bins on the opposite side of the object. The rays from the source to those bins spread out in a fan. Each bin records the intensity transmitted along its ray. In the idealized Beer–Lambert model, dividing the transmitted intensity by the unobstructed intensity and taking the negative logarithm gives the line integral of attenuation along that ray. This is why the processed measurements can be fitted with a linear forward operator in the image X [2].

The forward operator P_i. Treat X_{pq} as the constant attenuation within pixel (p, q). Let $\ell_{ij,pq}$ be the length in millimeters of ray j in view i inside that pixel. The CPU line projector used here applies the map

$$P_i \colon \mathbb{R}^{512\times 512} \longrightarrow \mathbb{R}^d, \qquad [P_i(X)]_j = \sum_{p,q} \ell_{ij,pq} X_{pq}, \qquad d = 956. \tag{1}$$

The weights are fixed by the calibrated source and detector geometry. A ray contributes only to pixels it crosses. ASTRA traces the rays to apply the operator, so the solver can evaluate a selected set of views directly [4].

Factorized data fitting. We model the image as $X = UV^\top$, with $U, V \in \mathbb{R}^{512\times 64}$, to restrict its rank to at most 64. All views of a slice constrain the same two factors. Adding views adds measurements while keeping the factor dimensions fixed.

Let $\mathcal{I}_{\text{train}}$ be the set of N training view indices, and let $y_i \in \mathbb{R}^d$ be the processed low-dose measurements for view i. Comparing those measurements with the predicted readings gives the unweighted least-squares loss

$$f(U, V) = \frac{1}{N} \sum_{i\in\mathcal{I}_{\text{train}}} q_i(U, V), \qquad q_i(U, V) = \frac{\|P_i(UV^\top) - y_i\|_2^2}{2d}. \tag{2}$$

Each view contributes all its detector bins. The loss is quadratic in the image X. As a function of the factors, it is quartic and nonconvex.

Constraints and the SFRAMES formulation. To control the scale of the factors, we cap each factor's operator norm, its largest singular value. We also require the factors to be entrywise nonnegative, which makes the reconstructed attenuation nonnegative. These requirements define

$$C = \{(U, V) : \ \|U\|_{\text{op}} \le R_U, \ \|V\|_{\text{op}} \le R_V\}, \tag{3}$$

$$D = \{(U, V) : \ U \ge 0, \ V \ge 0\}. \tag{4}$$

We encode nonnegativity by setting $g = \iota_D$, which is zero on D and $+\infty$ elsewhere. This indicator acts directly on the factors, so T is the identity. With $x = (U, V)$ and f given by (2), the reconstruction problem has the SFRAMES form

$$\min_{x\in C} \{f(x) + g(Tx)\}. \tag{5}$$

The component gradients are Lipschitz on the compact feasible region C. Global smoothness on the ambient factor space would require an extension outside a neighborhood of C. Small strictly positive factors lie in the interiors of both C and D.

Backprojection and stochastic gradients. The adjoint P_i^*, called *backprojection*, distributes each detector residual back to the pixels crossed by its ray, using the same weights $\ell_{ij,pq}$. Thus $P_i^*(P_i(X)-y_i)$ is the image-space gradient of the unnormalized half squared residual. The chain rule through $X = UV^\top$ then gives the two factor gradients.

At each SFRAMES update, draw a fresh uniform 64-element subset $B \subset \mathcal{I}_{\text{train}}$ without replacement within the batch and independently between updates. The gradient estimate is

$$E_B = \frac{1}{|B|d} \sum_{i \in B} P_i^*(P_i(UV^\top) - y_i), \qquad \widehat{\nabla f}_B(U,V) = (E_B V, E_B^\top U). \tag{6}$$

This estimate is unbiased conditional on the current iterate. One joint gradient evaluation uses one forward projection and one adjoint restricted to B.

2 Data and acquisition geometry

The inputs are measured fan-beam sinograms from 2DeteCT [2]. Mode 1 supplies low-dose observations for fitting and held-out-view diagnostics. The image reference for each slice is the published Mode 2 nonnegative least-squares reconstruction from a separate high-dose acquisition. The development slice is `04300`. The evaluation slices are `04531` and `04800`. The official Zenodo archives supplied slices `04300` and `04800`, and the DeepInverse single-slice mirror supplied `04531` [3]. Experiment data sidecars `data/slice*.json` record download sources and TIFF checksums.

A *sinogram* stacks the detector readings from all angular views. Each raw Mode 1 sinogram has 3,601 views and 1,912 detector cells. Discard the repeated last view and sum adjacent detector cells. Apply the same binning to the dark field, which measures the detector offset, and the mean of the two flat fields, which measure unobstructed illumination. If $S_i, D_{\text{dark}}, F_{\text{flat}}$ denote the resulting arrays, the pre-reversal measurement is

$$-\log\left(\max\left\{\frac{S_i - D_{\text{dark}}}{F_{\text{flat}} - D_{\text{dark}}}, 10^{-6}\right\}\right).$$

Reverse the detector axis to match the DeepInverse convention [3]. The recorded clipping count was zero for each of the three sinograms. Each processed acquisition has $3{,}600 \times 956$ entries.

We use the ASTRA 2.5.0 CPU `line_fanflat` projector in physical millimeters. The source-to-origin distance is 431.019989 mm, the origin-to-detector distance is 97.980499 mm, and the binned detector spacing is 0.1496 mm. The square image field has width 116.528143 mm and center $(-0.113797, -0.113797)$ mm. The angle of view i is $\pi - 2\pi i/3600$. Vertically flip each published 1024×1024 reference, divide by the native voxel width 0.113797 mm, and average 2×2 blocks to align with the 512×512 fitting grid.

Table 1: Exact view splits. Indices i range from 0 to 3599.

Role	Views	Selection rule
Smaller training set	360	$i \bmod 10 = 0$
Larger training set	1,080	$i \bmod 10 \in \{0, 3, 6\}$
Held-out validation	360	$i \bmod 10 = 9$
Calibration	120	$i \bmod 30 = 0$

Development runs use the smaller training set. Both training sizes are evaluated on each evaluation slice. Calibration views are a subset of both training sets and are disjoint from validation. The training split fixes which measured views define the objective. The optimizer then samples minibatches from that fixed split. Held-out loss measures how well the reconstruction predicts separate low-dose views of the same slice, while reference-image error compares the reconstructed attenuation values with the high-dose reconstruction.

3 Initialization and optimizer settings

Feasible calibration. Let P_J stack the 120 calibration views. Divide the positive backprojection $(P_J^* y_J)_+$ entrywise by $P_J^* P_J \mathbf{1}$, floored at 10^{-3} of its maximum. Initialize rank-64 nonnegative factors with an NNDSVD-style factorization of this image. Add 10^{-3} times each factor's RMS times independent uniform $[0, 1)$ entries. Fit a scalar amplitude by least squares on the calibration views with lower bound 10^{-6}, then multiply each factor by its square root. Denote the resulting factors by $U_{\mathrm{cal}}, V_{\mathrm{cal}}$. Set $R_U = 2\|U_{\mathrm{cal}}\|_{\mathrm{op}}$ and $R_V = 2\|V_{\mathrm{cal}}\|_{\mathrm{op}}$.

Estimate λ by ten power steps on $P_J^* P_J$, starting from a normalized all-ones image, and divide the resulting eigenvalue estimate by $120d$. The empirical block curvature scales are

$$L_U = \lambda \|V_{\mathrm{cal}}\|_{\mathrm{op}}^2, \qquad L_V = \lambda \|U_{\mathrm{cal}}\|_{\mathrm{op}}^2.$$

Both training sizes use the same calibration and caps within each slice. Table 2 gives the resulting values.

Table 2: Per-slice spectral caps, initial curvature scales, and smoothing prefactor.

Slice	R_U	R_V	L_U	L_V	μ_0
04300	3.95287036	3.94151485	0.19331899	0.19443450	5.14312016
04531	3.62203477	3.62670132	0.16367101	0.16325009	6.10981743
04800	4.03735408	4.03105829	0.20220242	0.20283452	4.93012733

SFRAMES. For $k \geq 0$, initialize momentum with the first sampled gradient and then use

$$m_k = (1 - \rho_k) m_{k-1} + \rho_k \widehat{\nabla f}_{B_k}(x_k), \qquad \gamma_k = \frac{\gamma_0}{(k+1)^{3/4}}, \qquad \rho_k = \frac{1}{(k+1)^{1/2}}, \qquad \mu_k = \frac{\mu_0}{(k+1)^{1/4}}.$$

The momentum formula applies for $k \geq 1$, and $\mu_0 = 1/\max(L_U, L_V)$. The exponents follow the SFRAMES manuscript [1]. Add the Moreau gradient after the momentum update and take the Frank–Wolfe step

$$s_k = \mathrm{LMO}_C\left(m_k + \frac{x_k - \Pi_D(x_k)}{\mu_k}\right), \qquad x_{k+1} = (1 - \gamma_k) x_k + \gamma_k s_k.$$

The projection Π_D clips negative entries. For a block direction $G = A \operatorname{diag}(\sigma) B^\top$, let A_+, B_+ contain the singular vectors corresponding to positive singular values. The exact spectral-ball LMO is $-R A_+ B_+^\top$, with optimal value $-R\|G\|_*$, where $\|\cdot\|_*$ is the nuclear norm. The FP64 thin SVD implementation retains singular values greater than $\varepsilon_{\mathrm{mach}} \max(\operatorname{shape} G) \sigma_{\max}(G)$. It returns zero for a zero direction. The contribution of discarded singular modes to the oracle gap is bounded by R times their sum.

Paired LMO start. The four-method comparison uses $U_{\text{cal}}, V_{\text{cal}}$ for all methods. The SFRAMES ablation retains the caps, curvature scales, rate settings, view splits, and work budgets. It starts with one LMO atom for each factor using independent standard Gaussian block directions. This places the start on the spectral-norm caps and outside D, and replaces the data-informed initial directions. Both variants start with empty momentum. The base seed is 110 in development and 201 in evaluation. Minibatches use NumPy's separate generator seeded with the base plus 100,000. The LMO directions use the base plus 200,000. Within each warm–LMO pair the minibatch stream is identical. The two evaluation view counts share the same LMO factors within each slice.

Comparators. SPRING-SARAH updates U and then V with the new U, using independent full refresh probability 1/16 for each block and 64-view recursive differences between refreshes. Both blocks begin with a full gradient at their respective query points [5]. ProxSARAH uses the joint factor gradient, with 256-view snapshots, 12 recursive updates of 64 views per snapshot, and relaxation 0.8 [6]. Each recursive difference evaluates the same sampled views at its two query points. PALM uses the full training set at each alternating block step [7]. Each proximal block step uses the Euclidean projection onto the intersection of its spectral ball and nonnegative orthant. The individual projections clip singular values at the radius and clip negative entries at zero, respectively. If a single-set projection is feasible for the other set it also solves the intersection problem. Otherwise Dykstra iterations combine the two projections with correction terms. A feasible candidate and a primal–dual gap control the numerical error. In exact arithmetic the distance from a feasible candidate to the true projection is bounded by the square root of twice this gap. The implementation includes a floating-point allowance and records floor-limited terminations. It rejects a proximal step when the numerical acceptance test fails within the iteration limit. The requested projection-distance tolerance is $10^{-8}/(k+1)^{1.1}$, with a check every five Dykstra iterations and at the first and last iterations. The limit is 2,000 iterations per factor projection.

Table 3: Development grids and frozen evaluation step rules. The rate multiplier is a.

Method	Candidate rates	Selected	Step rule
SFRAMES	0.05, 0.10, 0.25	0.10	$\gamma_0 = a,\ \rho_0 = 1$
SPRING-SARAH	0.30, 0.70	0.70	$(\eta_U, \eta_V) = (a/L_U, a/L_V)$
ProxSARAH	0.40, 0.90	0.90	$\eta = a/(L_U + L_V)$
PALM	0.60, 1.10	1.10	$(\eta_U, \eta_V) = (a/L_U, a/L_V)$

The selected rate minimizes repaired training loss at the last probe within the development budget among the completed runs. The table reports the practical pilot settings. The initialization ablation repeats all SFRAMES development rates and the frozen evaluation rate without selecting a new rate.

4 Work accounting and diagnostics

Each run stops after the first completed update that reaches its target of 5,760 forward views in development or 21,600 in evaluation. Save probes at iteration zero, after the first update, upon crossing multiples of 720 or 2,160 views respectively, and at the endpoint. Forward and backprojected views are recorded separately. Counts include refreshes and both evaluations in a SARAH difference.

Raw training loss is f at the actual iterate. Repaired training loss and held-out loss use a feasible reconstruction. The latter uses the normalization in (2) over the held-out set. Relative reference-image error is the Frobenius error of the repaired reconstruction divided by the norm of the aligned Mode 2 image. For the raw factors, the distance to D is the product Frobenius norm of their negative parts. Maximum violation is the maximum of zero, the two spectral-cap excesses, and the largest negative-entry magnitude in either block. For reporting, clip negative entries and scale the clipped block toward

zero when its operator norm exceeds its cap. Repair and diagnostic evaluations leave the optimizer state unchanged.

Elite used FP32 ASTRA forward and adjoint operations and FP64 factor arithmetic. Each worker used one BLAS thread on a distinct physical core. Optimizer timings include projections and oracle calls and exclude calibration, diagnostic probes, and checkpoint I/O. Times reflect concurrent arms on the same host.

5 Results

5.1 Four methods with a common feasible start

9 development runs and 16 evaluation runs completed. Each comparison uses the last saved probe within its budget. SFRAMES attained the lowest repaired training loss in 4 of 4 cases at the projection budget and in 4 of 4 at 100 optimizer seconds. Table 4 summarizes the view-budget comparison. Appendix A gives the saved trajectories.

Table 4: Medians at the last saved probe within 21,600 forward views and the same number of back-projected views.

Method	Raw training	Repaired training	Held-out	Reference error
SFRAMES	0.012915	0.012928	0.013084	35.88%
SPRING-SARAH	0.024481	0.024481	0.024504	59.12%
ProxSARAH	0.022315	0.022315	0.022347	55.81%
PALM	0.027301	0.027301	0.027329	62.59%

At 100 optimizer seconds, median repaired losses were 0.013078 for SFRAMES, 0.025225 for SPRING-SARAH, 0.023268 for ProxSARAH, 0.028121 for PALM. Evaluation runs reached their stopping budget in 109.7 to 120.4 optimizer seconds.

The proximal evaluation arms made 1,888 factor projections. Of these, 0 entered Dykstra iterations and 1,888 terminated at the numerical floor. Median recorded projection time was 0.56 seconds per arm. The largest reported numerical distance bound was 3.46×10^{-6}.

5.2 Paired SFRAMES initialization ablation

The Gaussian-direction spectral LMO starts are outside the nonnegative orthant. All 7 completed pairs matched work at each saved probe. The maximum observed spectral-cap excess for the signed-start runs was 3.55×10^{-15}. Table 5 compares their endpoints.

Table 5: Paired SFRAMES endpoints. The original feasible initialization has zero distance to D.

				Repaired loss		Final distance to D	
Slice	Views	γ_0	LMO initial D dist.	Warm	LMO	Warm	LMO
04300	360	0.05	31.7958	0.017685	2.697325	0.012621	6.933793
04300	360	0.10	31.7958	0.016005	0.034255	0.020554	0.033779
04300	360	0.25	31.7958	0.017977	0.018249	0.052620	0.056906
04531	360	0.10	29.1044	0.010998	0.011659	0.009147	0.009978
04531	1,080	0.10	29.1044	0.011116	0.011731	0.009076	0.009435
04800	360	0.10	32.3953	0.014495	0.015326	0.009099	0.010115
04800	1,080	0.10	32.3953	0.014648	0.015425	0.009178	0.010047

The change in repaired loss at the matched evaluation endpoints, computed as LMO-start loss divided by warm-start loss minus one, ranged from +5.31% to +6.01%.

The signed LMO atoms change both initial feasibility and the information available in the starting factors.

Scope. The pilot uses one development slice, two evaluation slices, and one evaluation seed. The four-method results use the same feasible warm start. The initialization ablation compares SFRAMES starts at frozen settings.

References

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[2] M. B. Kiss et al. *2DeteCT: A large 2D expandable, trainable, experimental Computed Tomography dataset for machine learning*. Scientific Data, 10, article 576, 2023. DOI.

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[4] ASTRA Toolbox. *2D Projectors*. Fan-beam line projector documentation.

[5] D. Driggs, J. Tang, J. Liang, M. E. Davies, and C.-B. Schönlieb. *A Stochastic Proximal Alternating Method for Nonsmooth and Nonconvex Optimization*. SIAM Journal on Imaging Sciences, 14(4), 1932–1970, 2021. DOI.

[6] N. H. Pham, L. M. Nguyen, D. T. Phan, and Q. Tran-Dinh. *ProxSARAH: An Efficient Algorithmic Framework for Stochastic Composite Nonconvex Optimization*. Journal of Machine Learning Research, 21(110), 1–48, 2020. Article.

[7] J. Bolte, S. Sabach, and M. Teboulle. *Proximal alternating linearized minimization for nonconvex and nonsmooth problems*. Mathematical Programming, 146, 459–494, 2014. DOI.

A Four-method comparison plots

The evaluation arms in these figures share the data-informed feasible start.

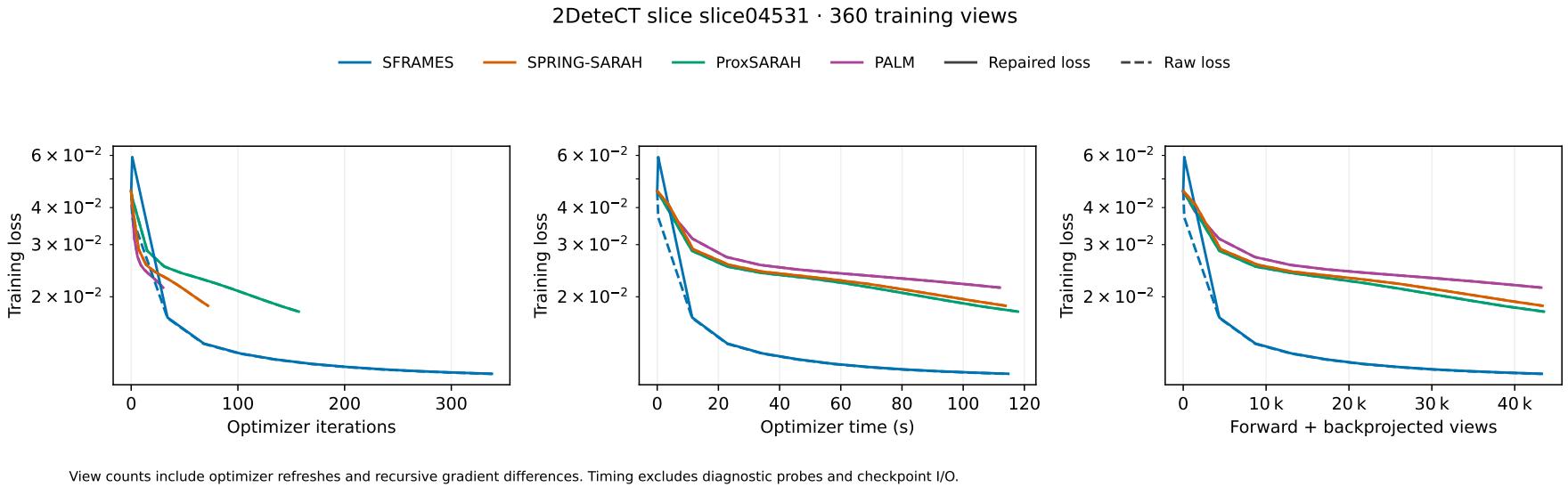


Figure 1: Slice 04531 with 360 training views. Raw and repaired training loss by update count, optimizer time, and projection work.

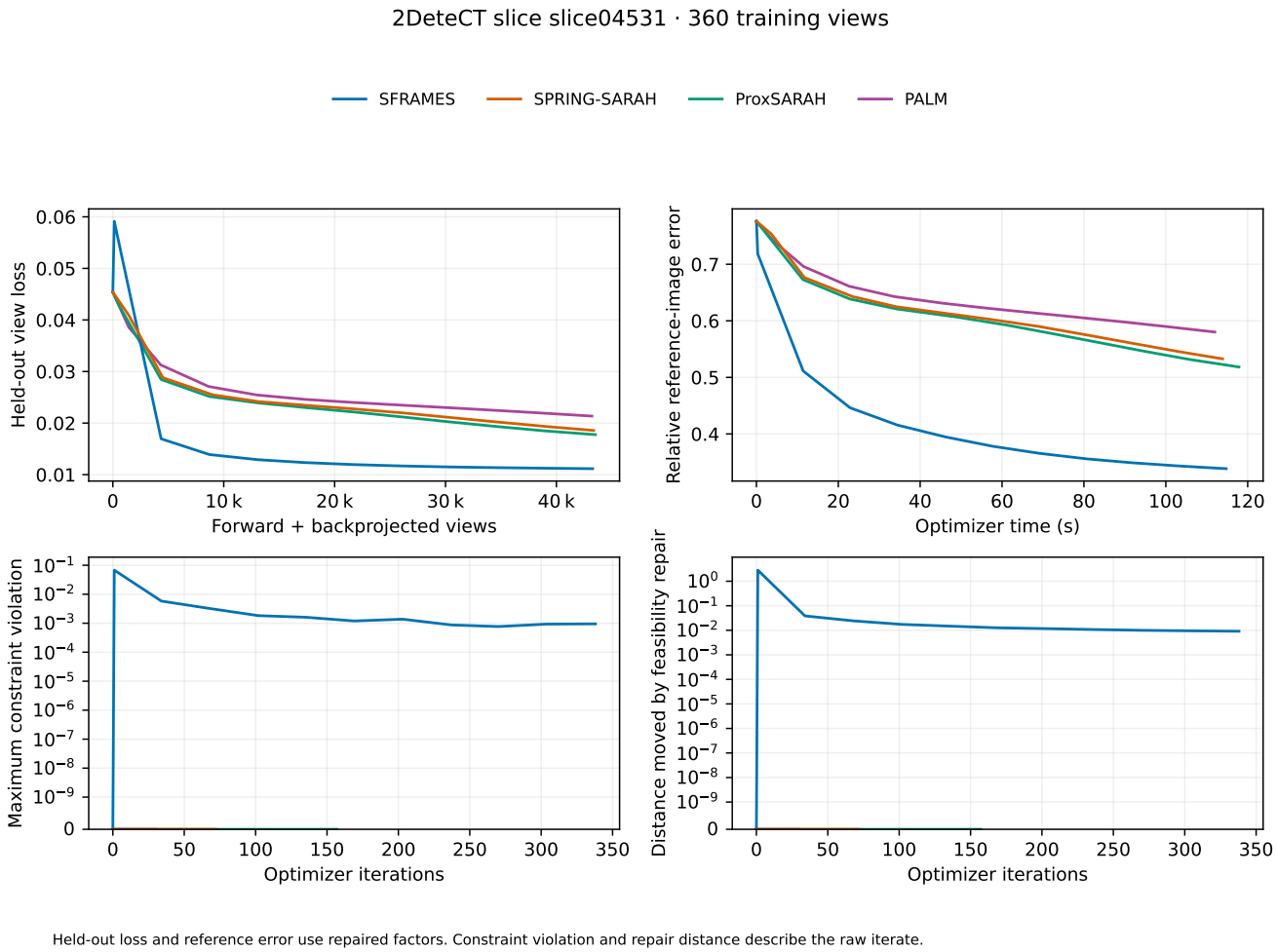


Figure 2: Slice 04531 with 360 training views. Held-out loss, reference-image error, and raw feasibility.

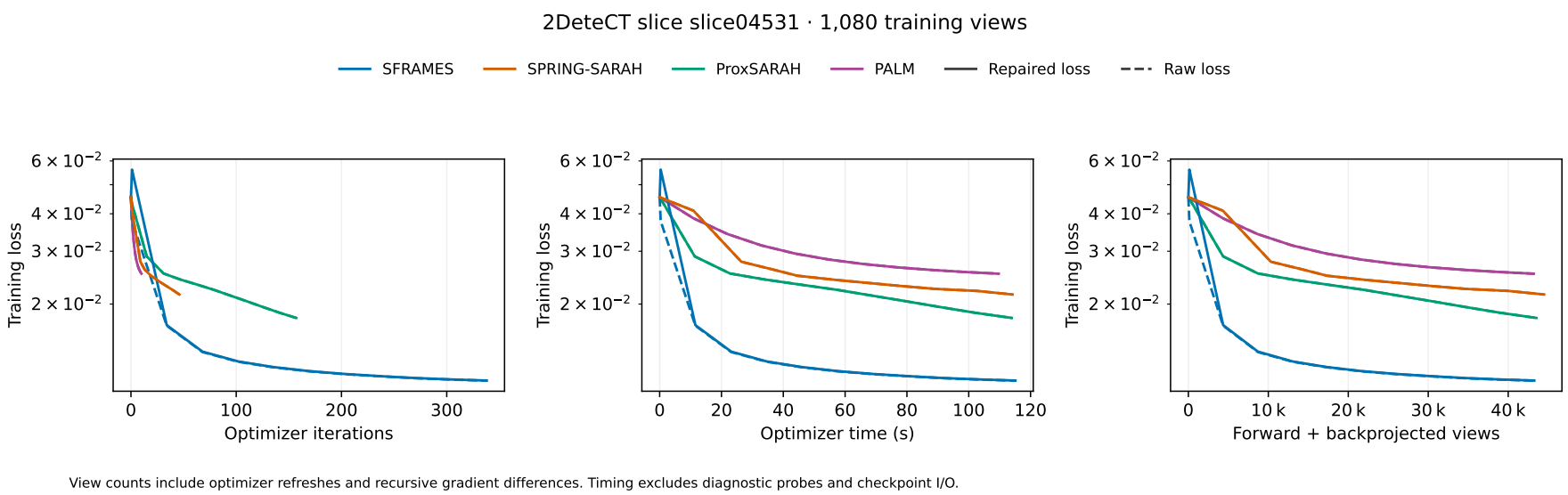


Figure 3: Slice 04531 with 1,080 training views. Raw and repaired training loss by update count, optimizer time, and projection work.

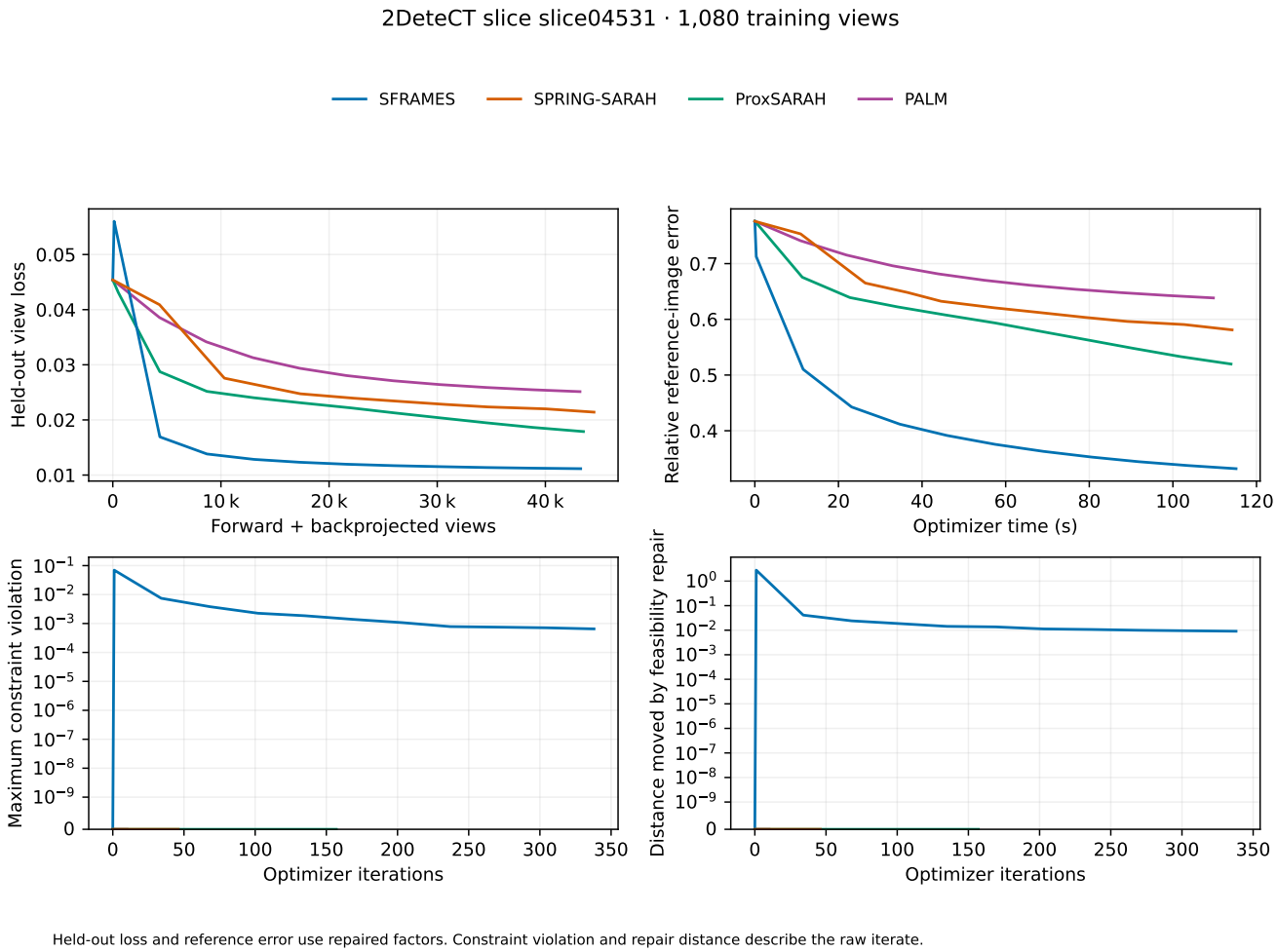


Figure 4: Slice 04531 with 1,080 training views. Held-out loss, reference-image error, and raw feasibility.

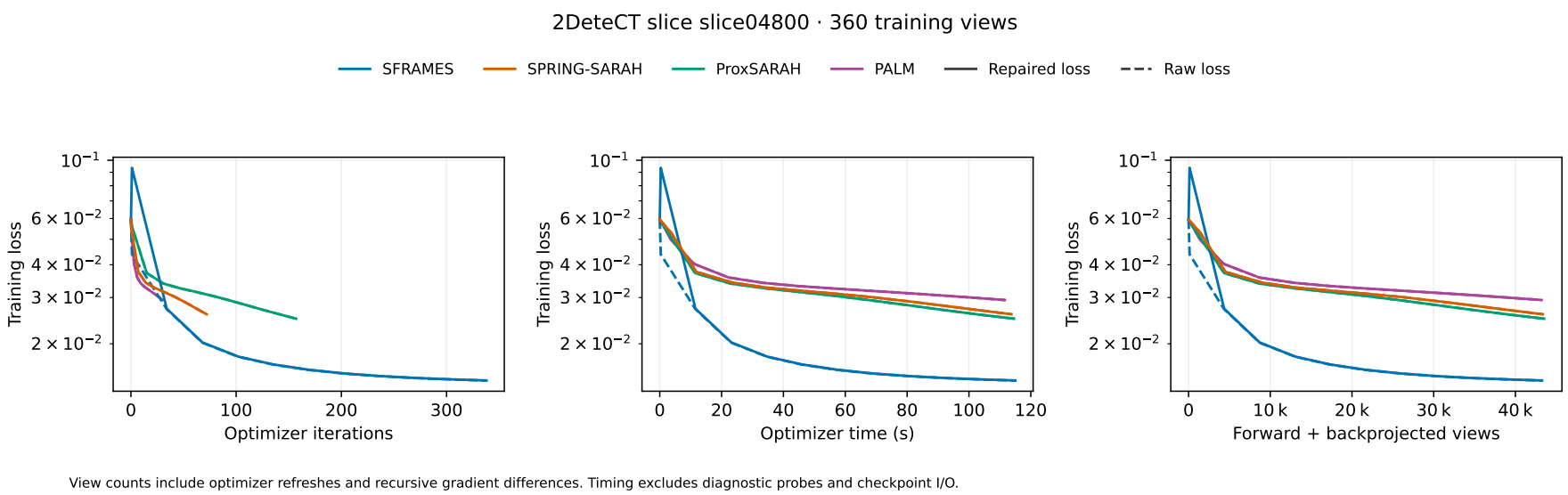


Figure 5: Slice 04800 with 360 training views. Raw and repaired training loss by update count, optimizer time, and projection work.

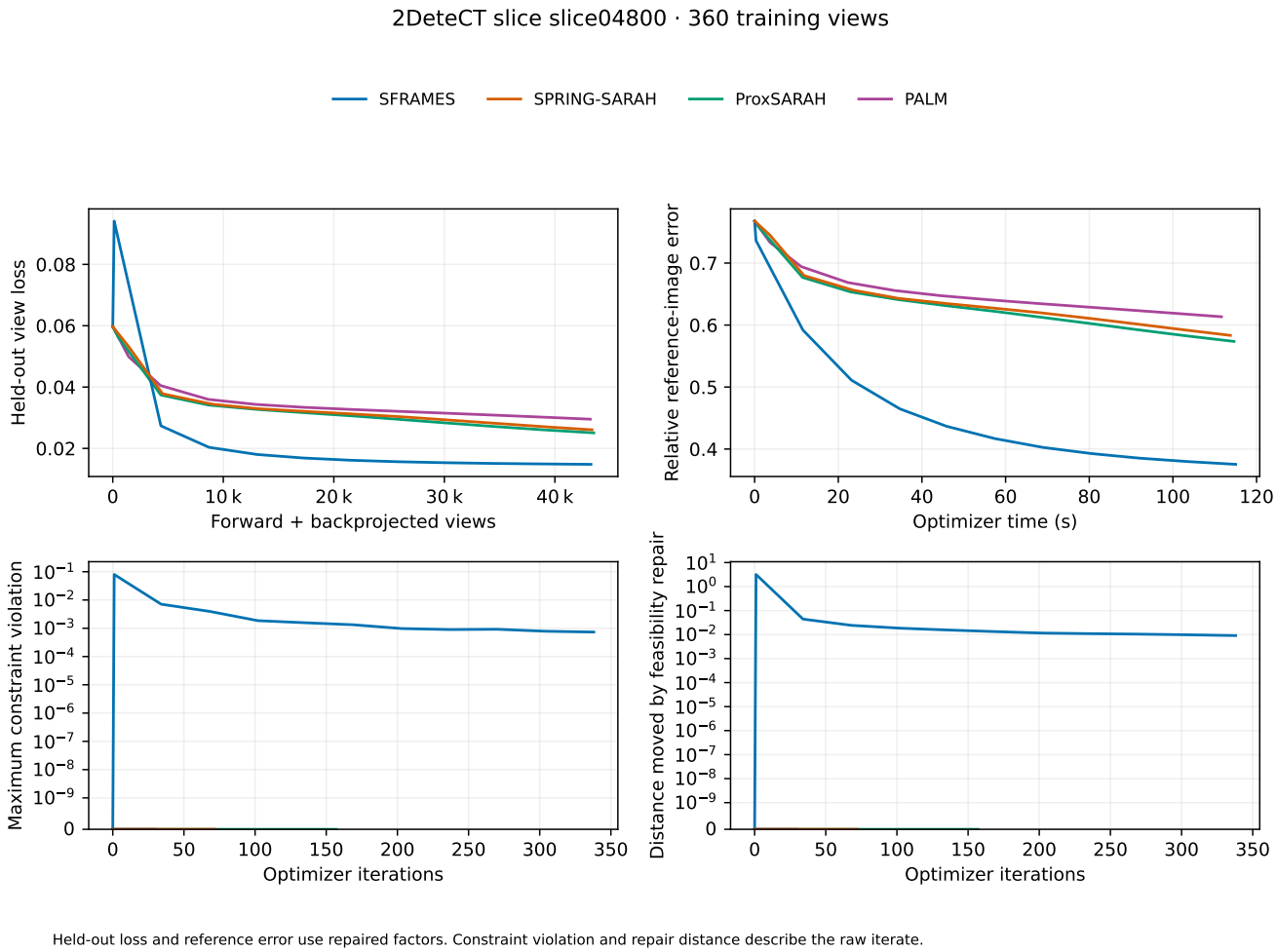


Figure 6: Slice 04800 with 360 training views. Held-out loss, reference-image error, and raw feasibility.

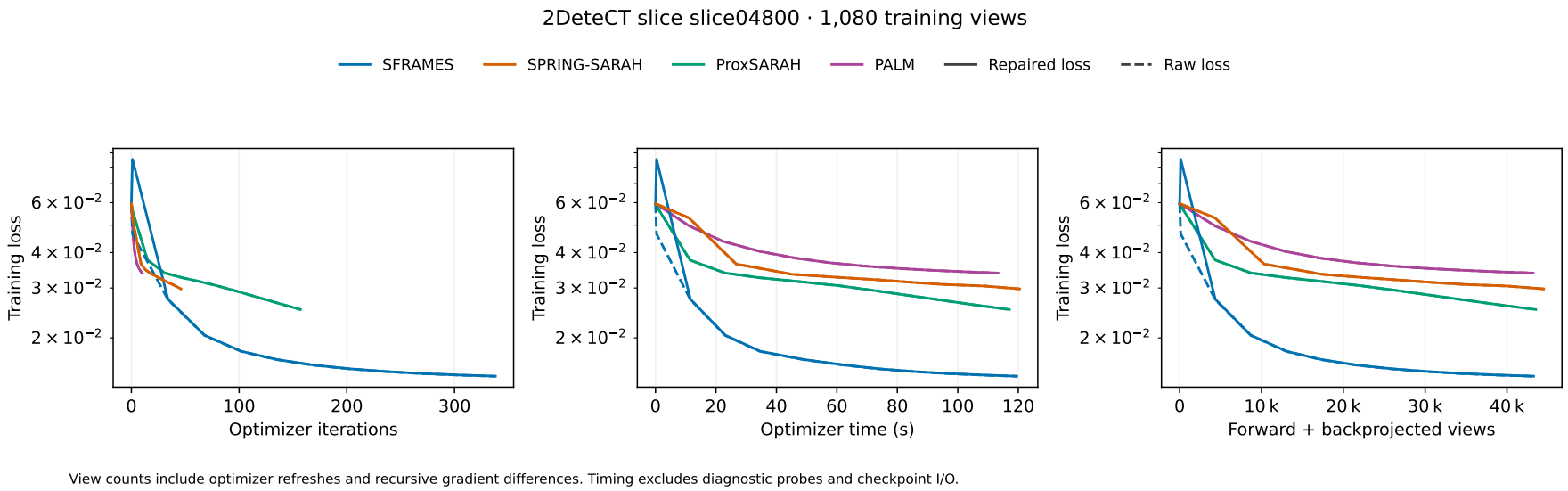


Figure 7: Slice 04800 with 1,080 training views. Raw and repaired training loss by update count, optimizer time, and projection work.

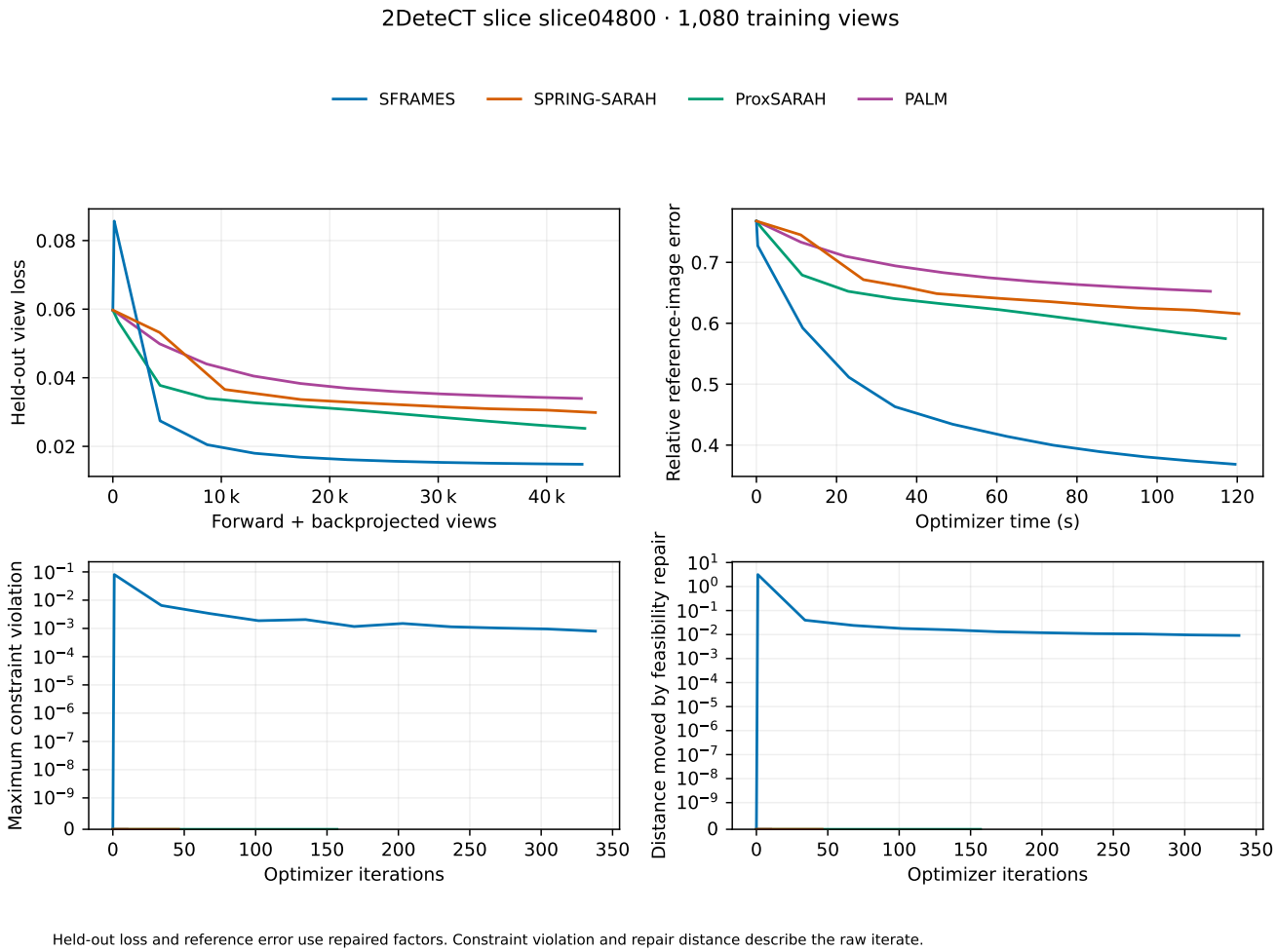


Figure 8: Slice 04800 with 1,080 training views. Held-out loss, reference-image error, and raw feasibility.